

Technology Trends Analysis using the Stack Overflow Developer Survey 2024 IBM Data Analyst Professional Certificate –CapstoneProject

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OUTLINE



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EXECUTIVE SUMMARY



This report presents an analysis of technology trends using data from the Stack Overflow Developer Survey, job postings, and web-scraped information. The objective is to identify current technology usage, emerging technology trends, and developer demographics to better understand the evolving software development landscape.

This report is intended for:

- Technology professionals seeking to develop in-demand technical skills.
- Students and aspiring developers planning their learning paths.
- Recruiters and hiring managers interested in labor market trends.
- Organizations making technology adoption and workforce planning decisions.

The findings provide data-driven insights into the technologies currently used by developers, those expected to grow in popularity, and the characteristics of today's developer community. These insights support informed decisionmaking regarding professional development, technology investments, and strategic workforce planning.



INTRODUCTION



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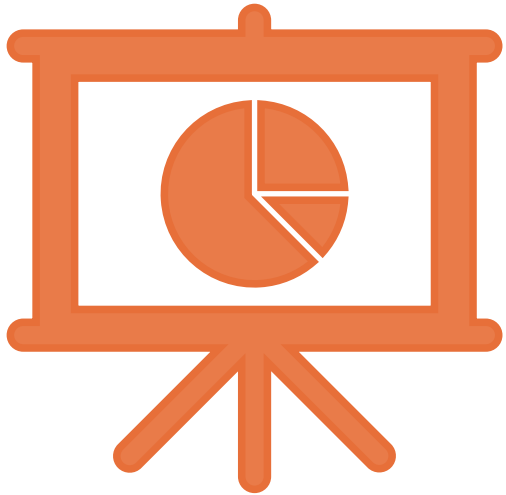
METHODOLOGY



Three primary data sources were used throughout this project: Stack Overflow Developer Survey for developer demographics, technology usage, and technology preferences. Job Posting API to identify current labor market demand for programming skills. Web Scraping to collect publicly available information on programming language popularity and technology trends. The datasets were collected using multiple techniques: CSV files provided by the Stack Overflow Developer Survey. API requests to retrieve job posting information. Web scraping techniques to extract publicly available technology data. These complementary sources provide a broader perspective on both developer preferences and industry demand. Several preprocessing techniques were applied to improve data quality before analysis: Removed duplicate records. Identified and handled missing values using appropriate imputation methods. Standardized data types and categorical variables. Removed outliers where appropriate. Performed exploratory data analysis (EDA). Created visualizations to identify trends and patterns. Developed an interactive dashboard to summarize the main findings.

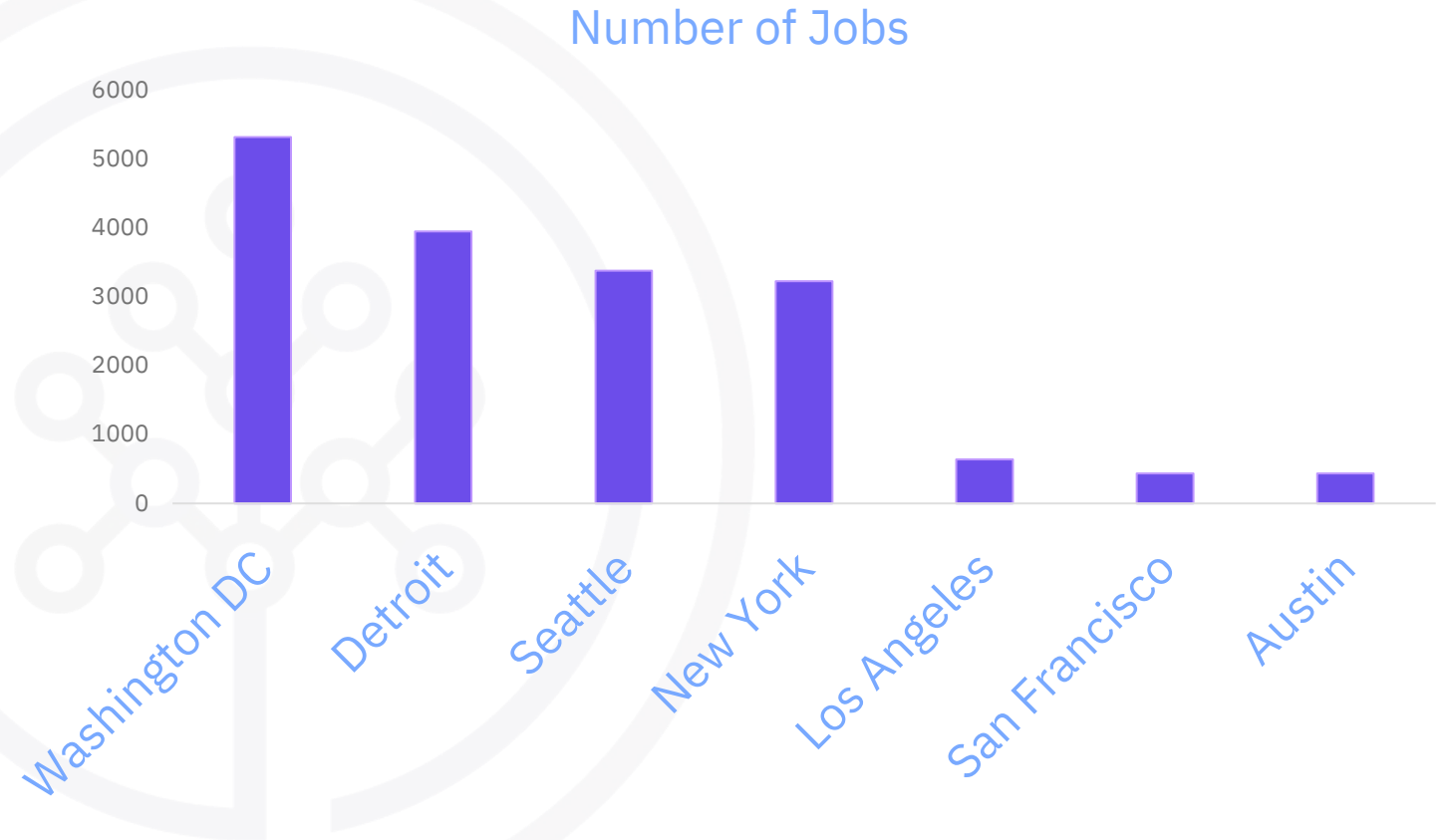


Results

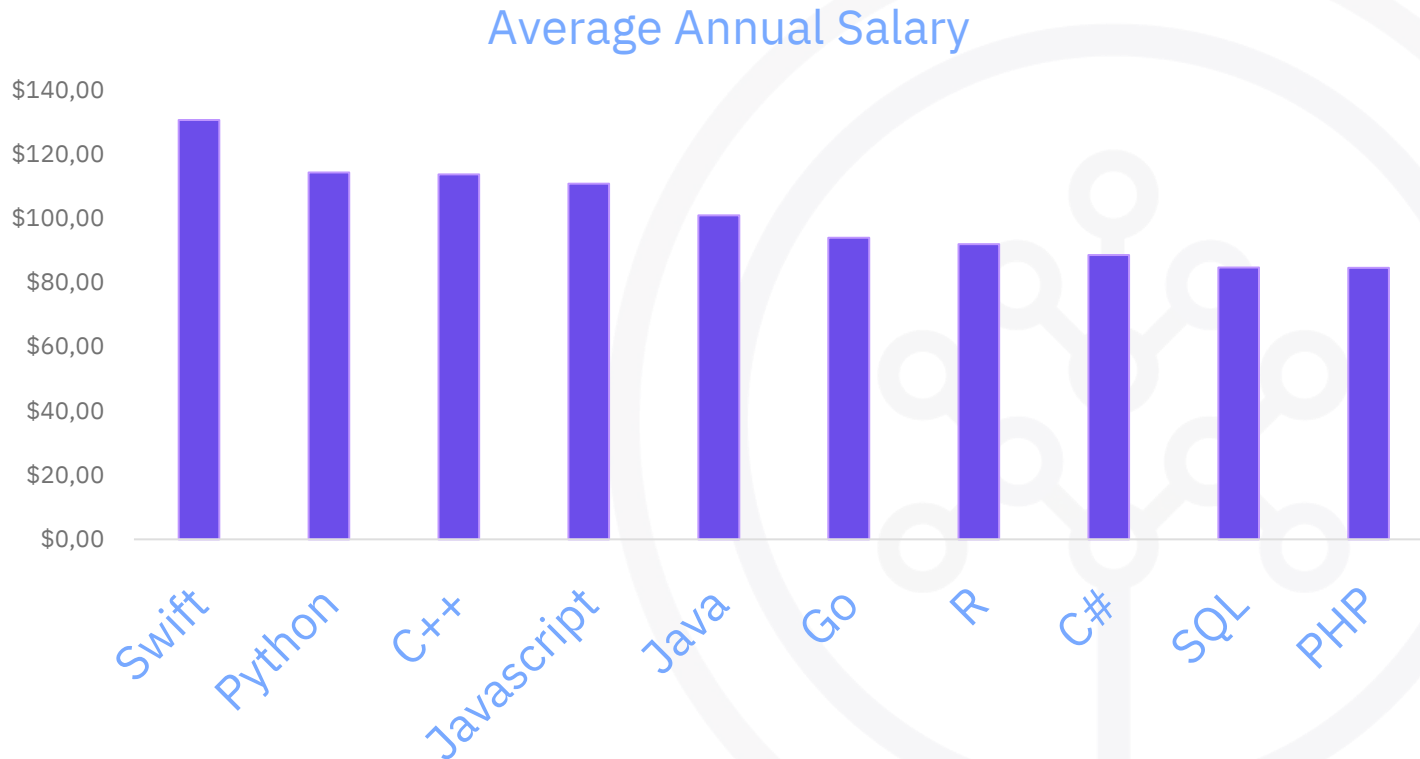


JOB POSTINGS

The chart presents the number of technology job postings across selected U.S. cities. It shows that major technology hubs, such as Washington DC, Detroit, and Seattle, have the highest demand for technology professionals, while cities such as Austin, San Francisco, Los Angeles, and New York also maintain significant employment opportunities. These results highlight the geographic concentration of technology jobs and the strong demand for skilled professionals in key metropolitan areas.



POPULAR LANGUAGES

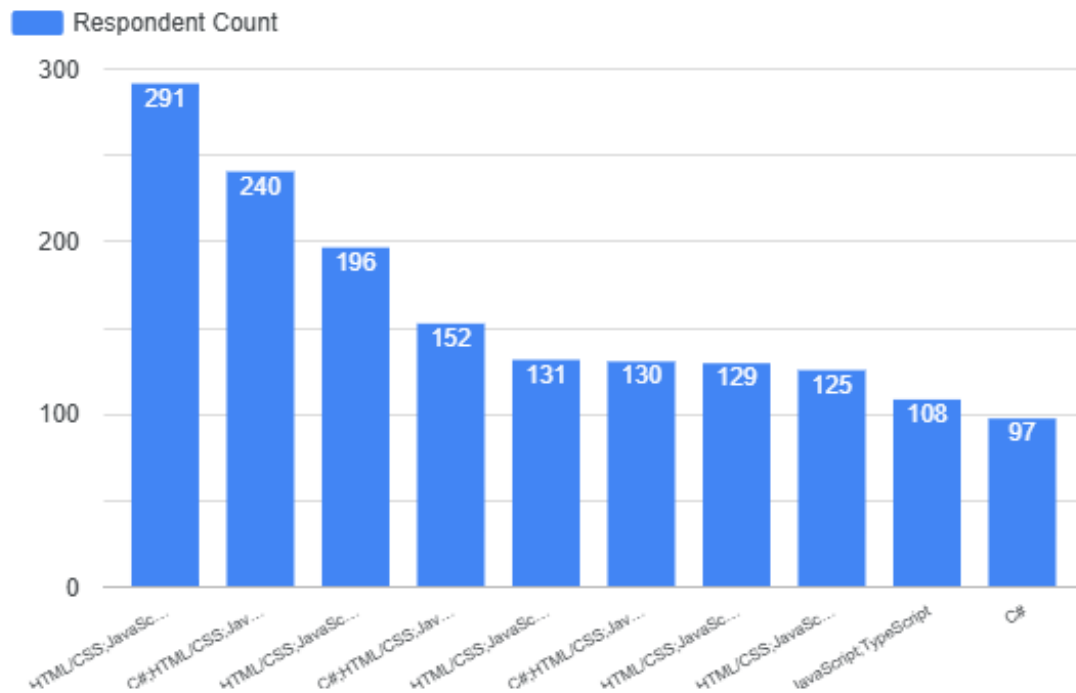


The chart compares the average annual salary associated with different programming languages. Swift has the highest average salary, followed by Python, C++, and JavaScript, while SQL and PHP are among the lowest-paying languages in this comparison. Overall, the results suggest that demand for specialized programming skills can be associated with higher compensation, although salary differences may also reflect factors such as industry, experience, and job role.

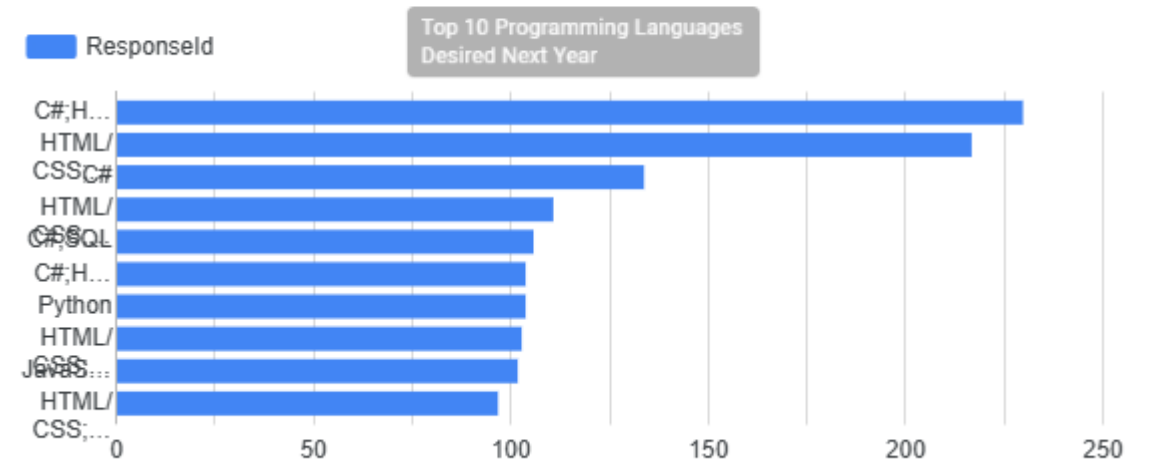


PROGRAMMING LANGUAGE TRENDS

Top 10 Programming Languages Used



Top 10 Programming Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

JavaScript, Python, and SQL are among the most widely used programming languages today. These languages also show strong future demand, indicating sustained industry relevance. Emerging technologies continue to gain adoption, but established languages remain the foundation of software development.

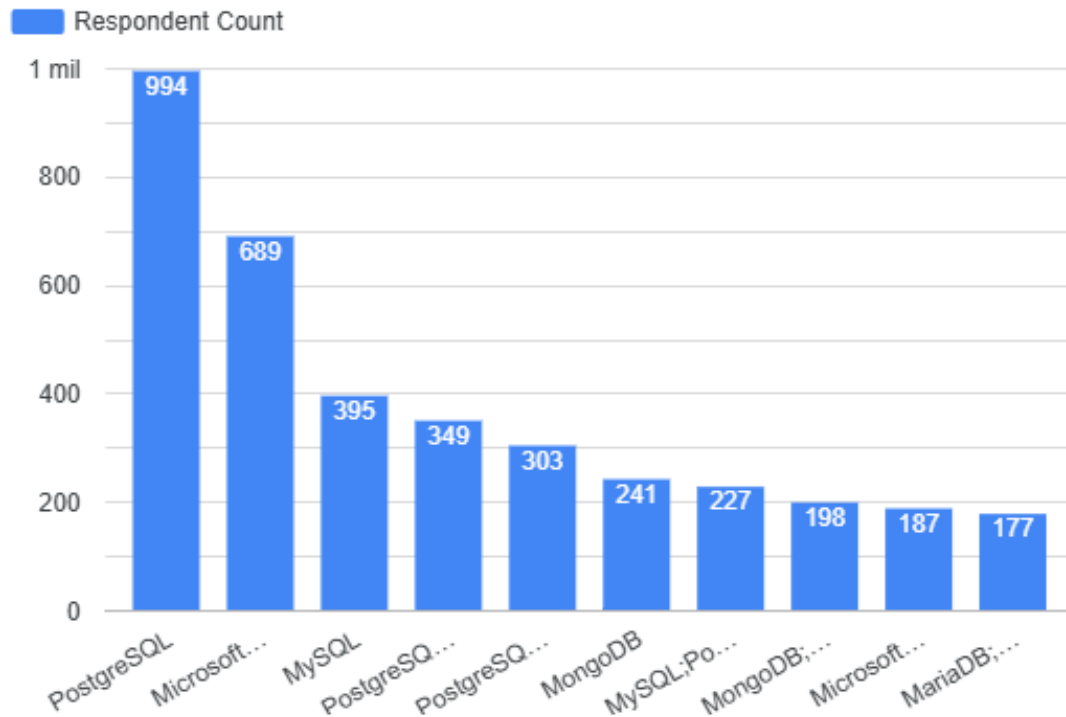
Implications

Developers should prioritize learning widely adopted languages to improve career opportunities. Organizations can reduce technology risk by investing in mature and broadly supported programming languages. Continuous upskilling is essential to keep pace with evolving technology trends and market demand.

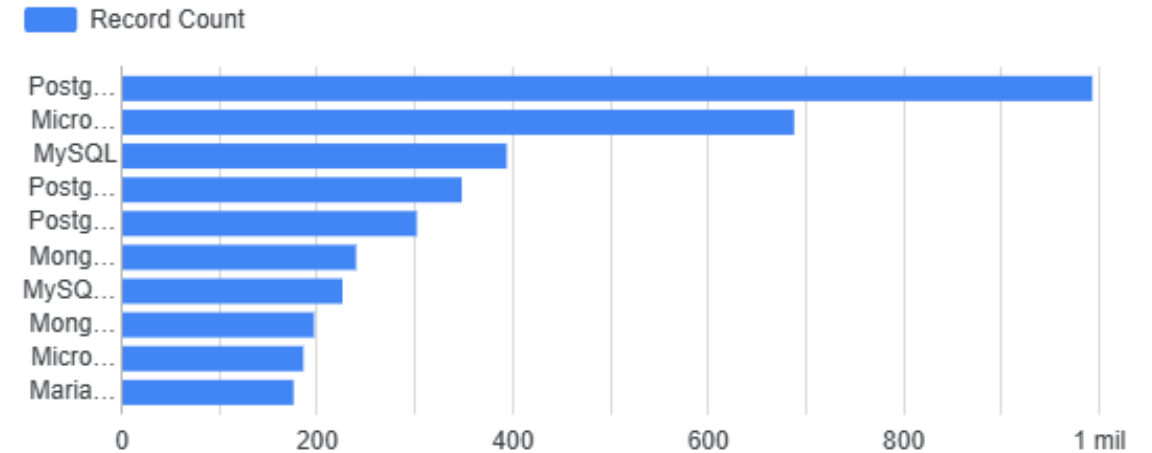


DATABASE TRENDS

Top 10 Databases Used



Top 10 Databases Desired Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

PostgreSQL is the most widely used database and remains the most desired technology for future adoption. Relational databases, including PostgreSQL, Microsoft SQL Server, and MySQL, continue to dominate the database landscape. NoSQL databases such as MongoDB maintain a strong presence, reflecting the growing need for flexible data storage solutions.

Implications

Professionals should develop strong SQL and PostgreSQL skills, as these technologies continue to lead both current usage and future demand. Organizations should maintain expertise in relational database management while also adopting NoSQL solutions when appropriate. Database professionals who combine relational and NoSQL knowledge will be better prepared for modern data engineering and analytics roles.



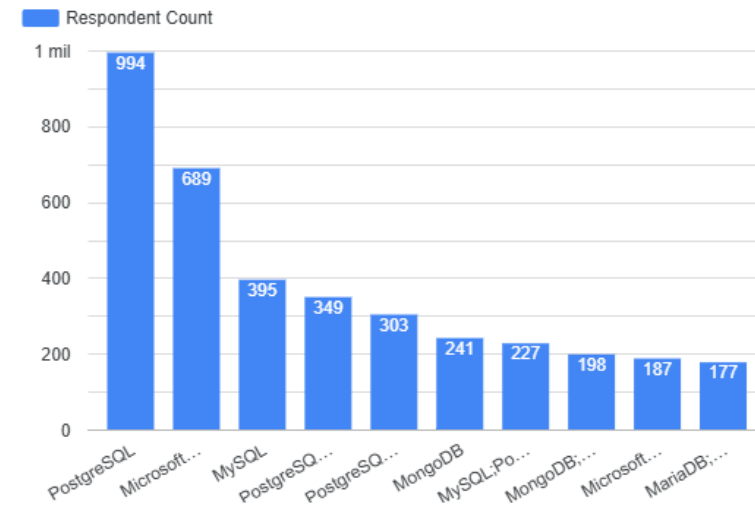
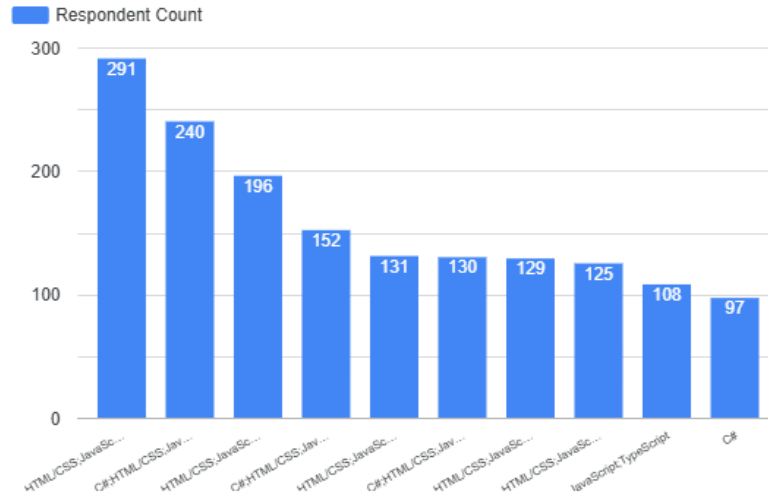
DASHBOARD



<https://datastudio.google.com/reporting/ffcfd000-33ba-4712-8063-d9416b5d3f0d>



DASHBOARD TAB 1: Current Technology Usage

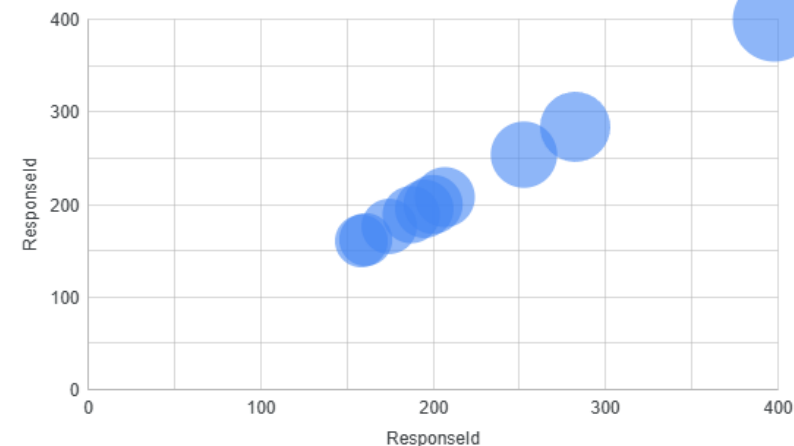


Top 10 Web Frameworks Used

Top Platforms Used



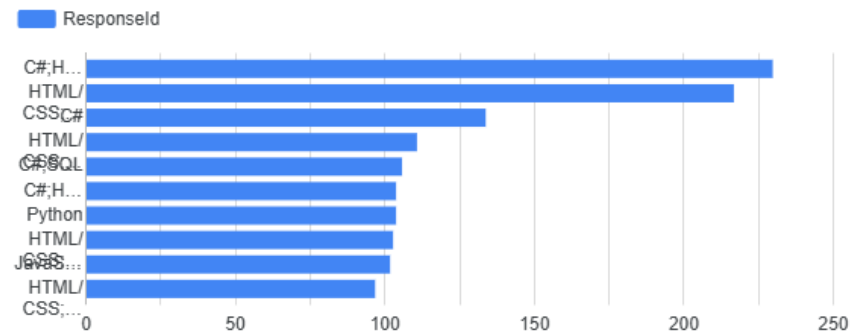
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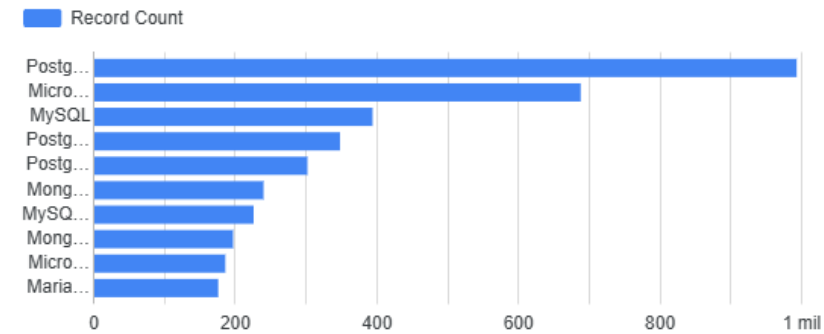
DASHBOARD TAB 2: Future Technology Trends

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Top 10 Programming Languages Desired Next Year



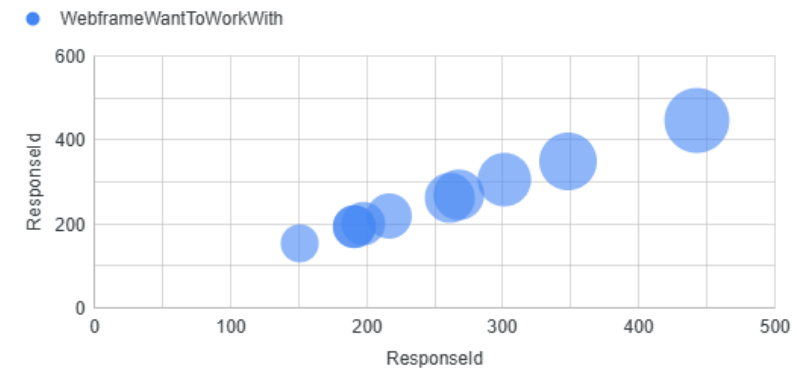
Top 10 Databases Desired Next Year



Top 10 Platforms Desired Next Year

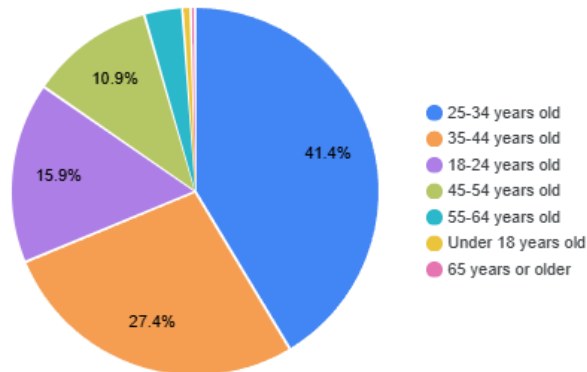


Top 10 Web Frameworks Desired Next Year

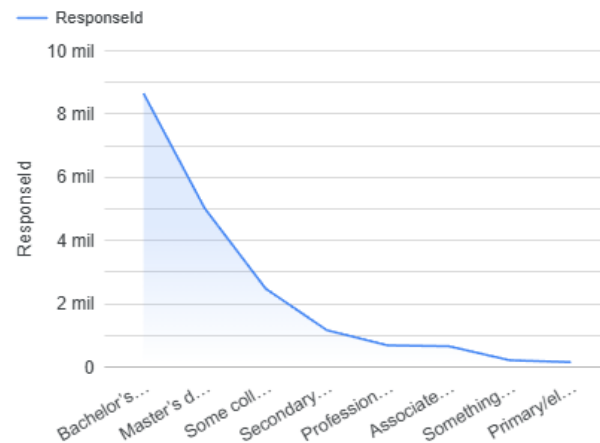


DASHBOARD TAB 3: Demographics

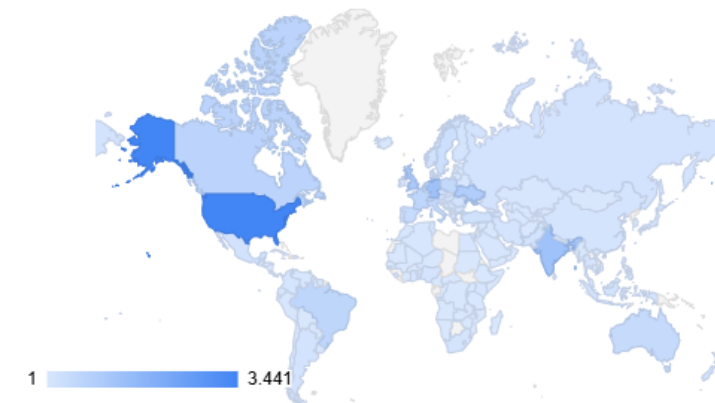
Respondent Distribution by Age



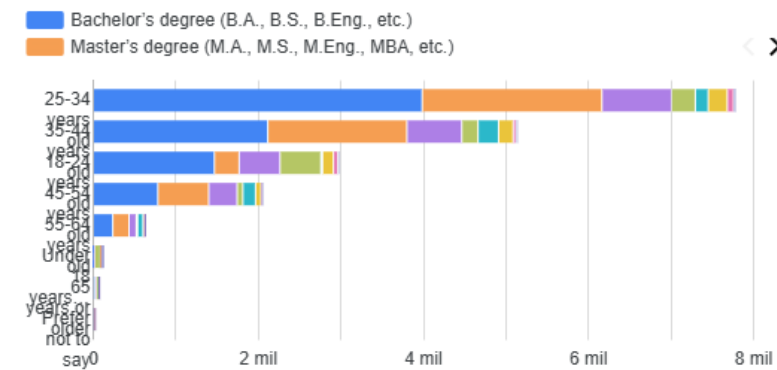
Respondent Distribution by Education Level



Respondent Count by Country



Respondent Count by Age and Education Level



DISCUSSION



The interactive dashboards provide a comprehensive view of the current technology landscape, future technology preferences, and the characteristics of the global developer community.

The Current Technology Usage dashboard shows that Python, JavaScript, SQL, PostgreSQL, Amazon Web Services (AWS), and Microsoft Azure are among the most widely adopted technologies, confirming their importance across software development and data-related roles.

The Future Technology Trends dashboard indicates that developers intend to continue investing in these technologies while showing increasing interest in modern cloud platforms, databases, and web frameworks. This suggests that organizations will continue prioritizing cloud computing, data engineering, and full-stack development skills.

Finally, the Demographics dashboard provides context for these trends by showing that most survey respondents are professionals in the early and middle stages of their careers, with diverse educational backgrounds and experience levels. This demonstrates that technology adoption is driven by a broad and active global developer community.

Overall, the dashboards indicate that Python, SQL, PostgreSQL, cloud platforms, and modern web technologies will remain fundamental skills for developers and organizations in the coming years.



OVERALL FINDINGS & IMPLICATIONS

Findings

Modern technologies such as Python, SQL, PostgreSQL, and cloud platforms consistently lead both current usage and future demand. The comparison between current and future trends indicates that core technologies will remain relevant while cloud computing and data-related skills continue to grow. The developer community is diverse but largely composed of early- and mid-career professionals, reflecting a rapidly evolving technology workforce.

Implications

Technology professionals should prioritize continuous learning in programming, databases, and cloud technologies to remain competitive. Organizations can use these insights to guide technology adoption, workforce planning, and employee training strategies. Data-driven decision making is essential for identifying technology trends and supporting long-term business and career development.



CONCLUSION



The analysis identified Python, SQL, PostgreSQL, and cloud platforms as the technologies with the strongest current adoption and future potential. Current and future technology trends indicate that these technologies will continue to play a central role in software development and data-driven industries. The combination of survey data, job market information, and dashboard visualizations provided a comprehensive understanding of the technology landscape. These findings can support professionals, organizations, and decision-makers in prioritizing technical skills, technology investments, and workforce development strategies.



APPENDIX

